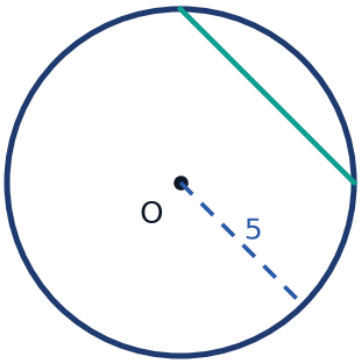
Geometry & Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

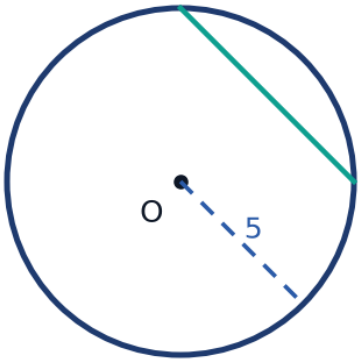
1 **[Foundational]** • **Target: Circles: chords and arcs**

Points A and B lie on a circle of radius 5 centered at O, with central angle $\angle AOB = 90^\circ$. Find the length of chord AB, using $AB = 2r\sin\left(\frac{\theta}{2}\right)$.



2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Points A and B lie on a circle of radius 5 centered at O, with central angle $\angle AOB = 90^\circ$. Find the length of chord AB, using $AB = 2r\sin\left(\frac{\theta}{2}\right)$.



3 **[Foundational]** • **Target: Distance and midpoint**

Find the distance between $(1, 2)$ and $(4, 6)$.

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4 [Foundational] • Target: Distance and midpoint

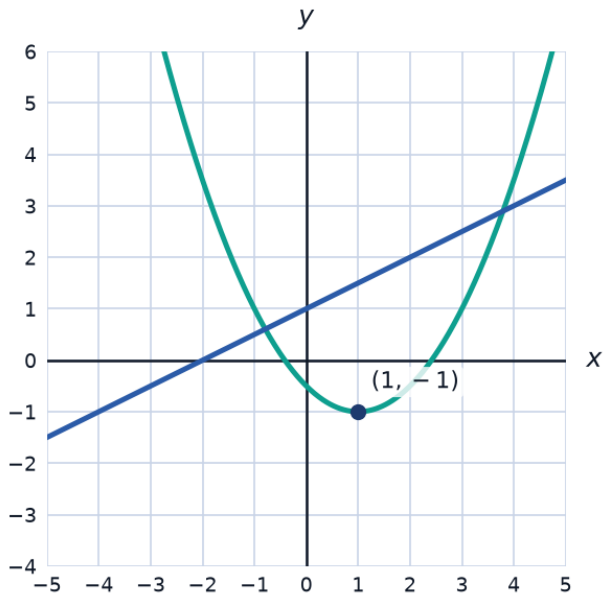
Find the midpoint of $(2, -3)$ and $(8, 5)$.

5 [Foundational] • Target: Distance and midpoint

Find the distance between $(-2, 1)$ and $(3, -11)$.

6 [Foundational] • Target: Distance and midpoint

Find the midpoint of $(-4, 7)$ and $(6, -3)$.



7 [Foundational] • Target: Gradient (slope)

Find the gradient of the line through $(1, 2)$ and $(5, 10)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Gradient (slope)

Find the gradient of the line through $(-3, 4)$ and $(2, -6)$.

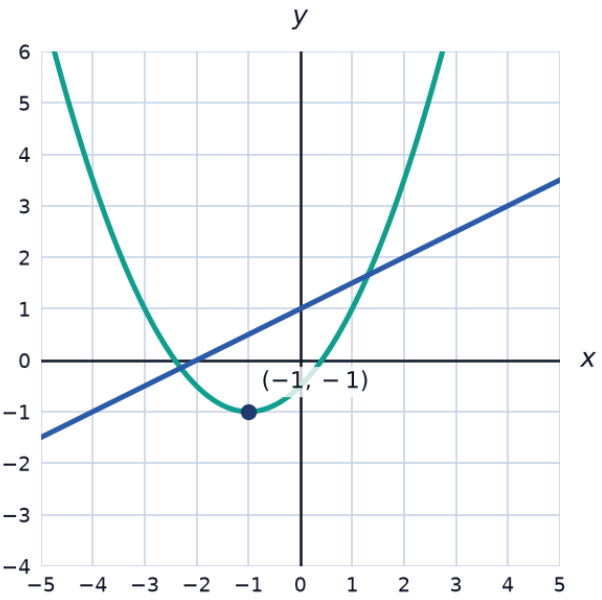
9 [Intermediate] • Target: Gradient (slope)

Find the gradient of the line through $(0, 0)$ and $(6, 9)$.

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10 [Intermediate] • Target: Gradient (slope)

State whether the line through $(1, 1)$ and $(4, 1)$ is horizontal, vertical, or neither.



11 [Intermediate] • Target: Equations of lines

Find the equation of the line with gradient 2 passing through $(3, 5)$, in the form $y = mx + c$.

12 [Intermediate] • Target: Equations of lines

Find the equation of the line through $(0, 4)$ and $(2, 10)$.

13 [Intermediate] • Target: Equations of lines

Find the x- and y-intercepts of the line $2x + 3y = 12$.

14 [Intermediate] • Target: Equations of lines

Write the equation of the line parallel to $y = 4x - 1$ passing through $(2, 9)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Parallel and perpendicular lines • Cross-Lesson: 9.01 Statistics & Probability

State the gradient of a line perpendicular to a line with gradient 5.

16 [Challenging] • Target: Parallel and perpendicular lines • Cross-Lesson: 1.01 Algebra & Equations

Determine whether the lines $y = 2x + 3$ and $y = 2x - 7$ are parallel, perpendicular, or neither.

17 [Challenging] • Target: Parallel and perpendicular lines • Cross-Lesson: 2.01 Functions & Transformations

Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 2$, passing through $(3, 4)$.

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18 [Challenging] • Target: The equation of a circle • Cross-Lesson: 3.01 Sequences & Series

Write the equation of a circle centered at the origin with radius 6.

19 [Challenging] • Target: The equation of a circle • Cross-Lesson: 4.01 Trigonometry

Write the equation of a circle centered at $(2, -3)$ with radius 5.

20 [Challenging] • Target: The equation of a circle • Cross-Lesson: 6.01 Limits & Derivatives

Find the radius and center of the circle $(x-1)^2 + (y+4)^2 = 49$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Distance and midpoint

Find the distance between the two points shown.

a. $\sqrt{5}$

b. 4

c. 3

d. 5

22 [Foundational] • Target: MCQ — Distance and midpoint

Find the midpoint of the segment joining the two points shown.

a. $(4, 3)$

b. $(2, 6)$

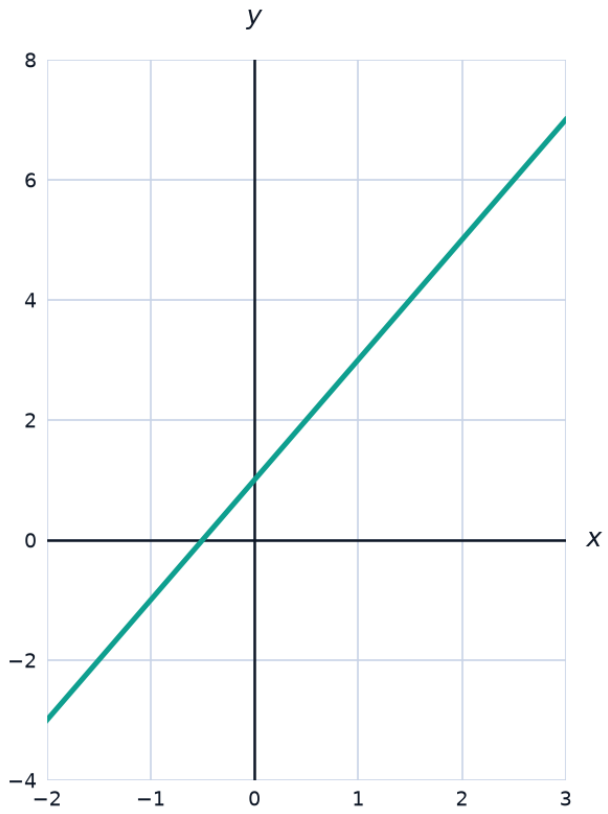
c. $(2, 3)$

d. $(8, 6)$

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23 [Foundational] • Target: MCQ — Equations of lines

The graph shows a line. Find its equation in the form $y = mx + c$ by reading the graph.



a. $y = \frac{1}{2}x + 1$

b. $y = 2x + 1$

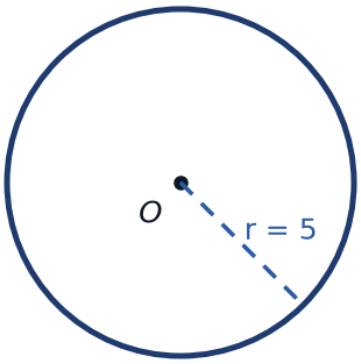
c. $y = x + 1$

d. $y = 2x - 1$

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24 [Foundational] • Target: MCQ — The equation of a circle

Write the equation of the circle shown, centered at the origin.



a. $x^2 + y^2 = 10$

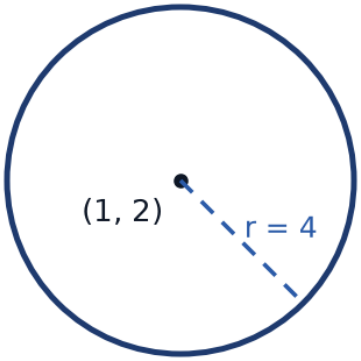
b. $x^2 + y^2 = 5$

c. $x^2 + y^2 = 20$

d. $x^2 + y^2 = 25$

25 [Foundational] • Target: MCQ — The equation of a circle

The circle shown is centered at $(1, 2)$ with the radius marked. Write its equation.



a. $(x - 1)^2 + (y - 2)^2 = 8$

b. $(x + 1)^2 + (y + 2)^2 = 16$

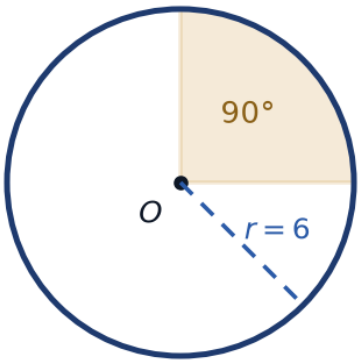
c. $(x - 1)^2 + (y - 2)^2 = 16$

d. $(x - 1)^2 + (y - 2)^2 = 4$

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26 [Foundational] • Target: MCQ — Circles: chords and arcs

Find the arc length of the 90° sector shown, in terms of π.



a. 12π

b. 3π

c. 1.5π

d. 6π

27 [Foundational] • Target: MCQ — Perimeter and area

A rectangle has length 15 and width 9, as shown. Find its perimeter and area.

a. $P = 135, A = 48$

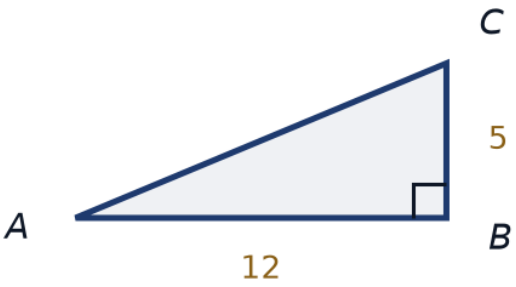
b. $P = 48, A = 135$

c. $P = 24, A = 135$

d. $P = 48, A = 24$

28 [Foundational] • Target: MCQ — Perimeter and area

A right triangle has legs 5 and 12, as shown. Find its area and hypotenuse.



a. $A = 60$, hyp $= 13$

b. $A = 30$, hyp $= 17$

c. $A = 30$, hyp $= 13$

d. $A = 30$, hyp $= 12$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: MCQ — Distance and midpoint**

Find the distance between $(2, 3)$ and $(6, 6)$.

a. 5

b. 25

c. $\sqrt{7}$

d. 7

30 **[Intermediate]** • **Target: MCQ — Gradient (slope)**

Find the gradient of the line through $(-2, 5)$ and $(3, -10)$.

a. $-\frac{1}{3}$

b. -15

c. -3

d. 3

31 **[Intermediate]** • **Target: MCQ — Equations of lines**

Find the x- and y-intercepts of the line $3x + 4y = 24$.

a. $x = 8, y = 6$

b. $x = 6, y = 8$

c. $x = 24, y = 24$

d. $x = 4, y = 3$

32 **[Intermediate]** • **Target: MCQ — Parallel and perpendicular lines**

Find the equation of the line perpendicular to $y = -\frac{1}{4}x + 3$, passing through $(4, 5)$.

a. $y = 4x - 5$

b. $y = 4x + 11$

c. $y = -\frac{1}{4}x + 6$

d. $y = 4x - 11$

33 **[Intermediate]** • **Target: MCQ — The equation of a circle**

Find the equation of the circle centered at the origin that passes through $(6, 8)$.

a. $x^2 + y^2 = 100$

b. $x^2 + y^2 = 10$

c. $x^2 + y^2 = 196$

d. $x^2 + y^2 = 48$

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34 [Intermediate] • Target: MCQ — Polygon angle sums

Find the measure of each interior angle of a regular hexagon.

a. 140°

b. 135°

c. 108°

d. 120°

35 [Intermediate] • Target: MCQ — Coordinate geometry: identifying shapes

Is the triangle with vertices $A(0, 0)$, $B(6, 0)$, $C(6, 4)$ a right triangle?

a. No

b. Cannot be determined

c. Yes

d. Only if it is also isosceles

36 [Intermediate] • Target: MCQ — The perpendicular bisector of a segment

A line segment has endpoints $(3, 6)$ and $(9, 6)$. Find the equation of its perpendicular bisector.

a. $x = 3$

b. $y = 6$

c. $y = 9$

d. $x = 6$

37 [Intermediate] • Target: MCQ — Circles: chords and arcs

Find the distance between $(3, 2)$ and $(-2, 14)$.

a. 15

b. 13

c. 17

d. 11

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Polygon angle sums • Cross-Lesson: 3.01 Sequences & Series

Find the midpoint of $(7, 6)$ and $(-9, 2)$.

a. $(0, 4)$

b. $(0, 5)$

c. $(-1, 5)$

d. $(-1, 4)$

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39 [Challenging] • Target: MCQ — Perimeter and area • Cross-Lesson: 4.01 Trigonometry

Find the gradient of the line through $(-3, 2)$ and $(-6, 3)$.

a. -3

b. $-\frac{1}{3}$

c. $\frac{1}{3}$

d. -1

40 [Challenging] • Target: MCQ — Coordinate geometry: identifying shapes • Cross-Lesson: 6.01 Limits & Derivatives

Find the equation of the line with gradient 4 passing through $(-4, 3)$, in the form $y = mx + c$.

a. $y = 4x + 21$

b. $y = 4x + 19$

c. $y = -4x + 19$

d. $y = 4x - 19$

41 [Challenging] • Target: MCQ — The perpendicular bisector of a segment • Cross-Lesson: 7.01 Integration & Applications

State the gradient of a line perpendicular to a line with gradient $\frac{4}{3}$.

a. $-\frac{1}{2}$

b. $\frac{4}{3}$

c. $-\frac{3}{4}$

d. $\frac{3}{4}$

42 [Challenging] • Target: MCQ — Word problems: geometry on a coordinate grid • Cross-Lesson: 8.01 Exponentials & Logarithms

Write the equation of a circle centered at $(-6, -2)$ with radius 3.

a. $(x + 6)^2 + (y + 2)^2 = 16$

b. $(x - 6)^2 + (y + 2)^2 = 9$

c. $(x + 6)^2 + (y - 2)^2 = 9$

d. $(x + 6)^2 + (y + 2)^2 = 9$

43 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Statistics & Probability

Find the measure of each interior angle of a regular dodecagon.

a. 30°

b. 155°

c. 150°

d. 145°

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44 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Algebra & Equations

A right triangle has legs 12 and 16. Find its area and hypotenuse.

a. Area : 96. Hypotenuse : 21

b. Area : 96. Hypotenuse : 20

c. Area : 108. Hypotenuse : 20

d. Area : 192. Hypotenuse : 20

45 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Functions & Transformations

Find the equation of the perpendicular bisector of the segment joining $(4, 5)$ and $(-2, 13)$.

a. $y - 1 = \frac{3}{4}(x - 9)$

b. $y - 9 = \frac{3}{4}(x - 1)$

c. $y - 9 = -\frac{3}{4}(x - 1)$

d. $y - 9 = -\frac{4}{3}(x - 1)$

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